THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $p(x) = 3x - 1$, then $p(p(2)) =$ 1

(a) 13 **(b)** 15 **(c)** 16 **(d)** 14

2. A plan costs ₹50 plus ₹15 per unit; another costs ₹170 plus ₹5 per unit. They cost the same for ___ units. 1

(a) 11 **(b)** 13 **(c)** 14 **(d)** 12

3. If $x = 7$ is the zero of $kx - 56$, then $k =$ 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

4. The graph of $y = 7$ (a constant polynomial) is: 1

(a) a horizontal line **(b)** a vertical line

(c) a point **(d)** a line through the origin

5. For the pattern with nth term $5n + 2$, the sum of the 1st and 12th terms is: 1

(a) 70 **(b)** 69 **(c)** 71 **(d)** 68

6. For which value of n is $12n + 7 = 271$? 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 24

7. The area (in square units) of the triangle formed by the line $y = 6x + 24$ and the two axes is: 1

(a) 49 **(b)** 50 **(c)** 47 **(d)** 48

8. The x-coordinate of the point where $y = 6x + 4$ and $y = -x + 25$ meet is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

9. **Assertion (A):** The slope of the line $y = -4x + 7$ is -4. 1

Reason (R): For the line $y = ax + b$ the slope is b.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The expression $x^2 - 5$ is a linear polynomial. 1
Reason (R): The degree of a linear polynomial is 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $p(x) = 2x - 7$, find $p(p(x))$ and its value at $x = 5$. 2

12. Find the area of the triangle formed with the axes by the line $y = 4x + 20$. 2

13. Plan A costs ₹90 plus ₹40 per unit; Plan B costs ₹330 plus ₹20 per unit. For how many units do both cost the same? 2

14. If $p(x) = 4x + 3$, find $p(p(x))$ and its value at $x = 1$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the point of intersection of $y = 4x + 8$ and $y = -2x - 4$. 3

16. A linear polynomial has $p(1) = 18$ and $p(3) = 30$. Find $p(x)$, its zero and $p(3)$. 3

17. The nth term of a pattern is $16n + 5$. Find the number of terms that are less than 149 and the sum of the first two terms. 3

OR

Find the point of intersection of $y = 6x + 1$ and $y = x + 21$. 3

18. A linear polynomial has $p(1) = -18$ and $p(3) = -12$. Find $p(x)$, its zero and $p(10)$. 3

OR

The nth term of a pattern is $12n - 2$. Find the number of terms that are less than 178 and the sum of the first two terms. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Seema saves money in her piggy bank every week: Rs 7 in week 1, Rs 12 in week 2, Rs 17 in week 3, Rs 22 in week 4, and so on, the amount increasing by the same sum each week. 5
(i) Find the common difference.
(ii) Write the amount t_n saved in week n as a polynomial in n.
(iii) In which week does she save Rs 87?
(iv) Can she save exactly Rs 100 in some week? Give a reason.

20. Two taxi services: A charges ₹20 plus ₹9 per km; B charges ₹80 plus ₹4 per km. Find the distance for which both charge equally, the common fare, and which is cheaper for 6 km. 5

OR

A line passes through $(1, -6)$ and $(5, 18)$. Find its slope, equation, intercepts and the area of the triangle it forms with the axes. 5

END OF QUESTION PAPER — Total: 40 Marks